Diversity in the Living World

Biology is the science of life forms and living processes, and the living world it studies holds an amazing diversity of organisms.

How early humans saw the living world

- the difference between inanimate matter and living organisms was easy for early man to perceive
- some inanimate matter was deified, wind, sea and fire among it, and so were some animals and plants
- what all these revered objects shared was the sense of awe or fear that they evoked

Description came much later

- describing living organisms, human beings included, began much later in human history
- societies that took an anthropocentric view of biology could register only limited progress
- systematic and monumental description of life forms brought in, out of necessity, detailed systems of identification, nomenclature and classification

The biggest spin off of those studies

- similarities are shared among living organisms both horizontally and vertically
- all present day living organisms are related to each other, and to all organisms that ever lived on this earth
- that revelation humbled man and led to cultural movements for conservation of biodiversity

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Ernst Mayr, 1904 to 2004

- born on 5 July 1904 in Kempten, Germany
- the Harvard University evolutionary biologist called The Darwin of the 20th century
- one of the 100 greatest scientists of all time
- died at the age of 100, in the year 2004

At Harvard

- joined Harvard's Faculty of Arts and Sciences in 1953
- retired in 1975, taking the title Alexander Agassiz Professor of Zoology Emeritus

What his research spanned

- a career of nearly 80 years
- ornithology, taxonomy and zoogeography
- evolution and systematics
- the history and philosophy of biology

What he is remembered for

- made the origin of species diversity the central question of evolutionary biology that it is today, almost single-handedly
- pioneered the currently accepted definition of a biological species

The triple crown of biology

- the Balzan Prize in 1983
- the International Prize for Biology in 1994
- the Crafoord Prize in 1999
- these three are widely regarded as biology's triple crown

What indeed is life?

- the range of living types is amazing, and the habitats extraordinary: cold mountains, deciduous forests, oceans, fresh water lakes, deserts, hot springs
- a galloping horse, migrating birds, the valley of flowers, an attacking shark, all evoke awe and a deep sense of wonder
- so does ecological conflict and cooperation within a population and among populations of a community, and the molecular traffic inside a cell

Two questions inside one

- a technical question: what is living, as opposed to the non-living
- a philosophical question: what the purpose of life is
- as scientists we do not attempt the second; the reflection here is on what is living

1.1 Diversity in the living world

Variety is everywhere you look

- a large variety of organisms is visible around us: potted plants, insects, birds, pets, other animals and plants
- several organisms cannot be seen with the naked eye, yet they are all around us
- widening the area of observation widens the range and variety seen
- a dense forest holds a much greater number and kind of living organisms

Species and biodiversity

- each different kind of plant, animal or organism represents a species
- the number of species known and described ranges between 1.7 and 1.8 million
- this is biodiversity, the number and types of organisms present on earth

1.2 Naming what we find

Why naming had to be standardised

- as new areas and even old ones are explored, new organisms are continuously identified
- we know the plants and animals of our own area by local names
- local names vary from place to place, even within one country, which would create confusion about which organism is meant
- so naming is standardised, and an organism is known by the same name all over the world

Two processes behind a name

- nomenclature, the standardised naming of living organisms
- identification, describing the organism correctly and knowing which organism the name attaches to
- naming is only possible once identification has been done

The international codes

- scientists have established procedures for assigning a scientific name to each known organism, acceptable to biologists all over the world
- for plants, the International Code for Botanical Nomenclature (ICBN)
- for animals, the International Code of Zoological Nomenclature (ICZN)
- the codes ensure each organism has only one name, that a description leads people anywhere to that same name, and that the name is not already used for another known organism

1.3 Binomial nomenclature

A name in two components

- each name has a generic name and a specific epithet
- providing a name with two components is binomial nomenclature
- given by Carolus Linnaeus and practised by biologists all over the world
- the two word format was found convenient

Mango as the worked example

- the scientific name of mango is written Mangifera indica
- Mangifera represents the genus
- indica is the particular species, the specific epithet

Universal rules, 1 to 3

- biological names are generally in Latin and written in italics, Latinised or derived from Latin whatever their origin
- the first word represents the genus, the second denotes the specific epithet
- both words, when handwritten, are separately underlined, or printed in italics, to indicate their Latin origin

1.4 Completing the binomial

Rule 4, and the author citation

- the first word, the genus, starts with a capital letter, while the specific epithet starts with a small letter, as in Mangifera indica
- the author's name appears after the specific epithet, at the end of the biological name, written in an abbreviated form
- Mangifera indica Linn. indicates that this species was first described by Linnaeus

1.5 Classification and taxa

Why we classify

- studying all the living organisms is nearly impossible, so a means had to be devised
- classification groups anything into convenient categories based on some easily observable characters
- naming a group calls up the characters we associate with it: dogs and not cats, Alsatians among dogs, mammals as animals with external ears and body hair, wheat and not rice

Taxa

- the scientific term for these convenient categories is taxa
- taxa can indicate categories at very different levels; plants is a taxa and wheat is a taxa
- animals, mammals and dogs are taxa at different levels, since a dog is a mammal and mammals are animals

1.6 Taxonomy and systematics

Taxonomy

- classifying all living organisms into different taxa on the basis of characteristics is taxonomy
- external and internal structure, along with the structure of the cell, the development process and ecological information, form the basis of modern taxonomic studies
- characterisation, identification, classification and nomenclature are the processes basic to taxonomy

How it began, and systematics

- taxonomy is not something new; human beings have always wanted to know more about the various kinds of organisms, particularly with reference to their own use
- early needs of food, clothing and shelter meant the earliest classifications were based on the uses of various organisms
- interest widened to the relationships among organisms, and that branch of study is systematics
- systematics is derived from the Latin word systema, the systematic arrangement of organisms

Systematics widens

- Linnaeus used Systema Naturae as the title of his publication
- the scope of systematics was later enlarged to include identification, nomenclature and classification
- systematics takes into account evolutionary relationships between organisms

1.7 Taxonomic categories

A hierarchy of steps

- classification is not a single step process; it involves a hierarchy of steps, each step representing a rank or category
- since a category is part of the overall taxonomic arrangement it is called a taxonomic category, and all the categories together constitute the taxonomic hierarchy
- each category, a unit of classification, represents a rank and is termed a taxon (plural: taxa)

Insects as the illustration

- insects are a group of organisms sharing common features such as three pairs of jointed legs
- they are recognisable concrete objects which can be classified, and so were given a rank
- groups represent category, category denotes rank, and each rank represents a unit of classification
- these categories are distinct biological entities, not merely morphological aggregates

The common categories

- kingdom, phylum or division (for plants), class, order, family, genus and species
- all organisms, in the plant and animal kingdoms alike, have species as the lowest category
- placing an organism requires knowledge of the characters of an individual or group, which shows up similarities and dissimilarities within the same kind of organism and against others

Species

- a group of individual organisms with fundamental similarities
- one species is told from another closely related species by distinct morphological differences
- in Mangifera indica, Solanum tuberosum (potato) and Panthera leo (lion), the words indica, tuberosum and leo are specific epithets, while Mangifera, Solanum and Panthera are genera, a higher level of taxon
- a genus may carry more than one specific epithet: Panthera also has tigris, and Solanum includes nigrum

Species, concluded

- Solanum also includes melongena
- human beings belong to the species sapiens, grouped in the genus Homo
- the scientific name for human being is therefore written Homo sapiens

Genus

- a group of related species having more characters in common than species of other genera
- genera are aggregates of closely related species
- potato and brinjal are two different species, both in the genus Solanum
- lion (Panthera leo), leopard (P. pardus) and tiger (P. tigris) are all species of Panthera, a genus that differs from Felis, which includes cats

Family

- a group of related genera, with still fewer similarities than genus and species
- families are characterised on both vegetative and reproductive features of plant species
- in plants, the genera Solanum, Petunia and Datura are placed in the family Solanaceae
- in animals, Panthera is put with Felis in Felidae, while cat and dog fall into two different families, Felidae and Canidae

Order

- species, genus and family rest on a number of similar characters, but order and the higher categories are identified on aggregates of characters
- an order is an assemblage of families exhibiting a few similar characters, fewer than the genera within a family share
- the plant families Convolvulaceae and Solanaceae fall in the order Polymoniales, mainly on floral characters
- the animal order Carnivora includes the families Felidae and Canidae

Class

- a class includes related orders
- the order Primata, comprising monkey, gorilla and gibbon, is placed in the class Mammalia, along with the order Carnivora, which includes tiger, cat and dog
- class Mammalia has other orders also

Phylum

- classes comprising animals such as fishes, amphibians, reptiles and birds, along with mammals, constitute the next higher category, phylum

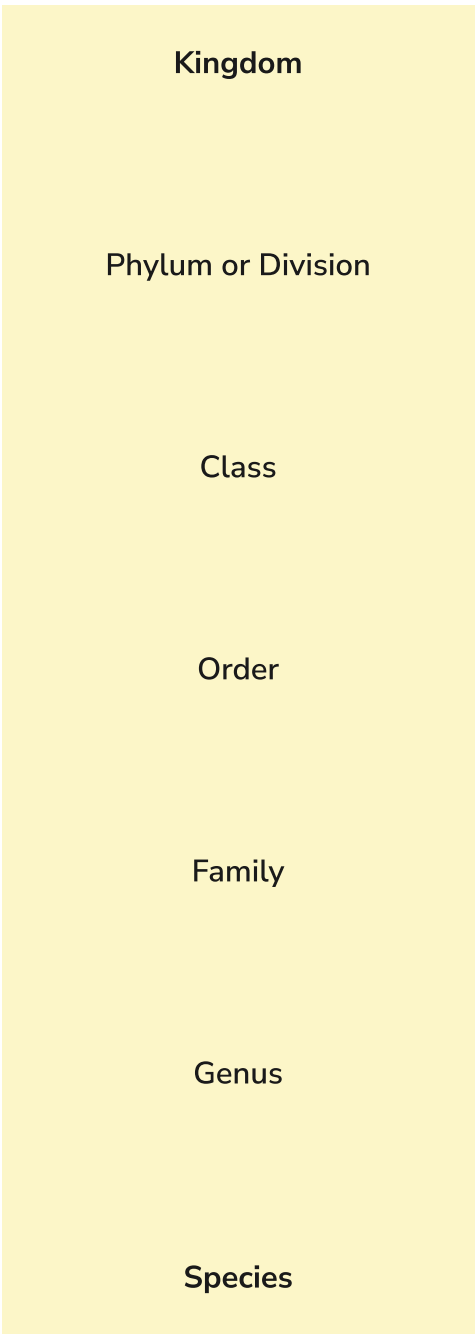


Figure 1.A Taxonomic categories showing hierarchial arrangement in ascending order

Phylum Chordata

- takes in fishes, amphibians, reptiles, birds and mammals
- common feature, the notochord
- common feature, the dorsal hollow neural system

Division

- the higher category on the plant side
- takes in classes that share only a few characters

1.8 Kingdom

Kingdom Animalia

- the highest category of all
- takes in animals from every phylum

Kingdom Plantae

- stands distinct from Animalia
- takes in plants from every division
- the two together, animal and plant kingdoms

The broad ranks

- species up to kingdom, the broad ranks, in Figure 1.A
- taxonomists have worked out sub-categories as well
- these give more sound and scientific placement of taxa

Climbing the hierarchy

- characters held in common go on falling
- the lower the taxon, the more its members share
- the higher the category, the harder to fix its relation to taxa at the same level
- so classification becomes more complex

Table 1.A places man, housefly, mango and wheat in the categories above.

Table 1.A **Organisms with their Taxonomic Categories**

Common Name	Biological Name	Genus	Family	Order	Class	Phylum/Division
Man	*Homo sapiens*	*Homo*	Hominidae	Primata	Mammalia	Chordata
Housefly	*Musca domestica*	*Musca*	Muscidae	Diptera	Insecta	Arthropoda
Mango	*Mangifera indica*	*Mangifera*	Anacardiaceae	Sapindales	Dicotyledonae	Angiospermae
Wheat	*Triticum aestivum*	*Triticum*	Poaceae	Poales	Monocotyledonae	Angiospermae

1.9 Summary

Variety, and the search for what defines life

- the living world is rich in variety; millions of plants and animals have been identified and described, but a large number remains unknown
- the range of organisms in size, colour, habitat, physiological and morphological features makes us seek the defining characteristics of living organisms

Taxonomy, and what it is good for

- to make the study of kinds and diversity possible, biologists evolved rules and principles for identification, nomenclature and classification
- the branch of knowledge dealing with these is taxonomy
- taxonomic studies of species of plants and animals are useful in agriculture, forestry and industry, and generally for knowing our bio-resources and their diversity

Naming and placing an organism

- the basics of taxonomy, identification, naming and classification, are universally evolved under international codes
- on resemblances and distinct differences, each organism is identified and assigned a correct scientific name of two words, as per the binomial system of nomenclature
- an organism occupies a place or position in the system of classification
- the many categories or ranks are referred to as taxonomic categories or taxa, and all of them together constitute a taxonomic hierarchy

Exercises

1 For what reason are living organisms classified?
2 What makes systems of classification change from time to time?
3 Which criteria would you pick to sort the people you meet often?
4 What does identifying individuals and populations teach us?
5 Of these two forms of the scientific name of mango, which one is written correctly: Mangifera Indica, or Mangifera indica?
6 State what a taxon is, and give examples of taxa at different levels of the hierarchy.
7 Which of these is the correct sequence of taxonomical categories? (a) Species, Order, Phylum, Kingdom (b) Genus, Species, Order, Kingdom (c) Species, Genus, Order, Phylum
8 Gather the meanings currently accepted for the word species, and discuss with your teacher what species means for higher plants and animals on one hand, and for bacteria on the other.
9 Define and understand these terms: (i) Phylum (ii) Class (iii) Family (iv) Order (v) Genus
10 Illustrate the taxonomical hierarchy, using a plant and an animal as examples.